FLATNESS OF A QUASI-COHERENT MODULE UNDER A FLAT MORPHISM OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays the pointwise equivalence and then states the global consequence; the proof establishes the displayed equivalence and the consequence follows by surjectivity.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Flatness at a point for modules on algebraic spaces. Lemmas lemma-flat-at-point and lemma-flat-at-point-etale-local-rings, used at proof lines 6524-6527. Reduce the sheaf statement to a statement about etale local rings.
- The stalk of a pullback of a quasi-coherent sheaf. Properties of Spaces, Lemma spaces-properties-lemma-stalk-pullback-quasi-coherent, used at proof lines 6527-6529. Identifies the stalk of f^*G at the geometric point with a tensor product.
- Flatness descends in the presence of the flat local ring map. Algebra, Lemma algebra-lemma-flatness-descends-more-general, used at proof lines 6535-6536. Is the algebraic statement the reduction yields.
- Flatness at a point is tested on geometric points. used at proof lines 6521-6523. Allows the argument to be carried out at one geometric point.

Source reference. Original lemma 0584: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6767–6767. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05VU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7563–7563. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04NG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6079–6079. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 05VQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7559–7559. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$, $g : Y \to Z$ be morphisms of algebraic spaces over S. Let $\mathcal{G}$ be a quasi-coherent sheaf on Y. Let $x \in |X|$ with image $y \in |Y|$. If f is flat at x, then*

$$\mathcal{G} \text{ flat over } Z \text{ at } y \Leftrightarrow f^*\mathcal{G} \text{ flat over } Z \text{ at } x.$$

In particular: If f is surjective and flat, then $\mathcal{G}$ is flat over Z, if and only if $f^*\mathcal{G}$ is flat over Z.

Proof. Pick a geometric point $\overline{x}$ of X and denote $\overline{y}$ the image in Y and $\overline{z}$ the image in Z. Via the characterization of flatness in Lemmas 05VU and 04NG and the description of the stalk of $f^*\mathcal{G}$ at $\overline{x}$ of Properties of Spaces, Lemma 05VQ the lemma reduces to a purely algebraic question on the local ring maps $\mathcal{O}_{Z,\overline{z}} \to \mathcal{O}_{Y,\overline{y}} \to \mathcal{O}_{X,\overline{x}}$ and the module $\mathcal{G}_{\overline{y}}$. This algebraic statement is Algebra, Lemma 0584. □